

WASP-12b

R-0211

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250103 · created 2026-07-17T00:13:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460678.7697 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.067 +/- 0.025
Transit depth (Rp/Rs)^2	0.45 +/- 0.33 [%]
Orbital Inclination (inc)	77.96 +/- 2.82
Airmass coefficient 1 (a1)	0.7142 +/- 0.0077
Airmass coefficient 2 (a2)	0.331 +/- 0.068
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#2 - [608, 75]
Optimal Aperture	4.13
Optimal Annulus	5.5
Transit Duration (day)	0.0726 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.067 ± 0.025 vs 0.1178 (deviation 1.74σ)
Duration fit vs published	0.0726 d vs 0.125 d (deviation 5.55σ)
Mid-time O–C	-1.4 min
Depth SNR	2.3
Residual scatter	1.08 %

Light curve

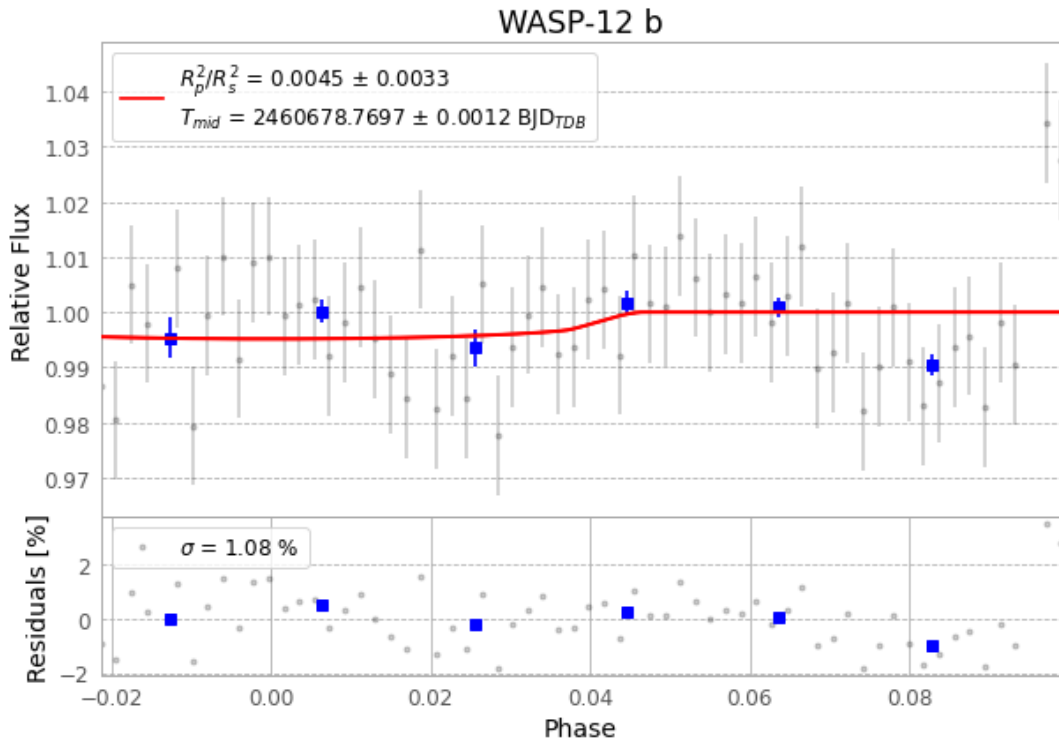


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-02.png (SHA-256 aeb2f703635ad80a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 261], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5881a15d5cb13a42...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

64 FITS frames (42 MB), WASP-12250103054515.FITS → WASP-12250103085415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250103.log (c207586d9572...), FinalLightCurve_WASP-12 b_2025-01-02.png (aeb2f703635a...), FinalLightCurve_WASP-12 b_2025-01-02.csv (d7e8f0361fca...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:13Z.